LeSuccess Academy

Learn . Educate . Succeed | Official Course Syllabus

Data Engineering

Duration: 3-6 Months | Mode: HYBRID | 100% Placement Support

Detailed Curriculum & Module Outline:

Module 1: Data Engineering Foundations

Where data engineering sits between sources and analysts, and the shape of a modern data platfo

Module 2: Advanced SQL

Window functions, CTEs, query plans and the optimisation habits a warehouse bill depends on

Module 3: Python for Data Engineering

File and API ingestion, Pandas, error handling and scripts written to be rerun safely

Module 4: Databases and Data Modelling

Normalisation, star and snowflake schemas, fact and dimension tables, slowly changing dimension

Module 5: ETL and ELT Pipeline Design

Extract, transform and load patterns, idempotency, incremental loads and handling late data

Module 6: Apache Spark

RDDs, DataFrames, partitions, joins and shuffles, processing datasets too large for one machine

Module 7: Hadoop and Distributed Storage

HDFS, MapReduce concepts and where the older ecosystem still turns up in enterprise stacks

Module 8: Workflow Orchestration with Airflow

DAGs, dependencies, retries, backfills and scheduling work that must not silently skip

Module 9: Cloud Data Platforms

S3 as a data lake, Redshift and Snowflake as warehouses, and moving data between them

Module 10: Streaming Data Basics

Kafka concepts, topics and consumers, and when a stream beats a nightly batch

Module 11: Data Quality and Governance

Validation, tests, lineage, access control and being able to prove a number is right

Module 12: Capstone Pipeline Project

A scheduled, monitored pipeline from raw source to warehouse tables an analyst can query